BNPL Machine Learning Project Report

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Course: CS 133

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Abstract

Buy now, pay later (BNPL) is a financial tool that has seen a rapid increase in usage over the last 10 years. The core idea is simple, make a purchase now and pay it off over time. BNPL is however risky for both the provider, and for the user as with everything that involves money. Our dataset is composed of 2 million transactions from a Nigerian BNPL app. The goal of this project is to see if we can understand and predict if a person is able to successfully pay off their BNPL purchase. Our visualization strategy revolved around the default rate and the credit scores of customers, as we felt those were the most impactful features for answering our main questions. Credit score emerged as the dominant predictor of repayment behavior, with high-risk customers defaulting at a 33.1% rate compared to just 0.1% for low-risk customers. Our logistic regression and gradient boosting models successfully classified 90-day default risk and estimated credit scores, enabling customer segmentation into three distinct risk tiers.

Background / Problem Definition

Dataset

The dataset used in this project is the electricsheep Africa/nigerian-banking-bnpl dataset, sourced from HuggingFace. It contains 2,000,000 rows and 16 columns representing individual BNPL transactions originating from Nigeria. The dataset was selected because it captures a real-world, high-stakes financial problem in an emerging market where BNPL adoption is growing rapidly, making default risk prediction both practically important and analytically interesting.

Key features include:

- transaction_id, customer_id — unique identifiers
- purchase_date, first_payment_due — datetime fields already parsed correctly
- merchant_category, merchant_name — categorical purchase context (groceries, electronics, fashion, furniture, other)
- customer_state — geographic location across 37 Nigerian states/territories
- principal_ngn — loan amount in Nigerian Naira (float64)
- interest_rate_monthly — monthly interest rate; ~30% of rows are 0 (legitimate interest-free deals)
- tenor_days, num_installments — loan duration and repayment structure
- provider — BNPL provider (FairMoney, Branch, Renmoney, Carbon, PayLater)
- credit_score — integer score ranging from ~300 to 850
- first_time_customer — boolean flag
- default_30d, default_90d — boolean repayment outcome targets

The dataset is clean, zero null values across all 2 million rows. The main data quality consideration is the 600,377 rows (~30%) where interest_rate_monthly equals zero, which represent legitimate interest-free promotional BNPL deals rather than data errors. These were retained to be considered as a potential ML feature.

Example of preprocessing we did on the data:

```
# Make a working copy
viz_df = df.copy()

# Convert date columns if present
date_cols = ["purchase_date", "first_payment_due"]
for col in date_cols:
    if col in viz_df.columns:
        viz_df[col] = pd.to_datetime(viz_df[col], errors="coerce")

# Convert target columns to numeric if needed
for col in ["default_30d", "default_90d", "credit_score", "principal_ngn", "annual_income_ngn"]:
    if col in viz_df.columns:
        viz_df[col] = pd.to_numeric(viz_df[col], errors="coerce")

viz_df.head()
```

Problem Statement

The main question we're trying to answer is: based on what we know about a borrower, such as their credit score, loan size, merchant, location, and provider, can we predict whether they'll default?

This has real-world importance. If a BNPL company can identify high-risk borrowers early, it can adjust interest rates, limit loan amounts, or decline applications altogether. In Nigeria, this is especially important because many people are underbanked and don't have a formal credit history. Getting these predictions right could help expand access to financial services while managing risk.

Analytical Questions:

https://github.com/CS133-DataVisualization/term-project-dgim/blob/main/BNPL_Data_Visualization_duc.ipynb

1. How are loan amounts distributed across merchant categories, and does what you're buying affect how much you borrow?
2. What's the relationship between credit score, loan amount, and default risk, and is there a clear score threshold where defaults start to drop?
3. Do borrowers with lower credit scores default more, and does the BNPL provider make a difference within each credit tier?
4. How does the 90-day default rate change across credit scores for different providers, and can this be explored interactively?
5. Do default rates vary across Nigerian states, and is there a geographic pattern in repayment behavior?

Real World Relevance

BNPL has grown quickly, with companies like Carbon, FairMoney, and Branch offering short-term credit to people who may not qualify for traditional bank loans. This helps improve financial inclusion, but it also means lenders are taking on more risk without the usual safeguards. A strong default prediction model can help providers lend more responsibly and may also allow reliable borrowers to access better rates.

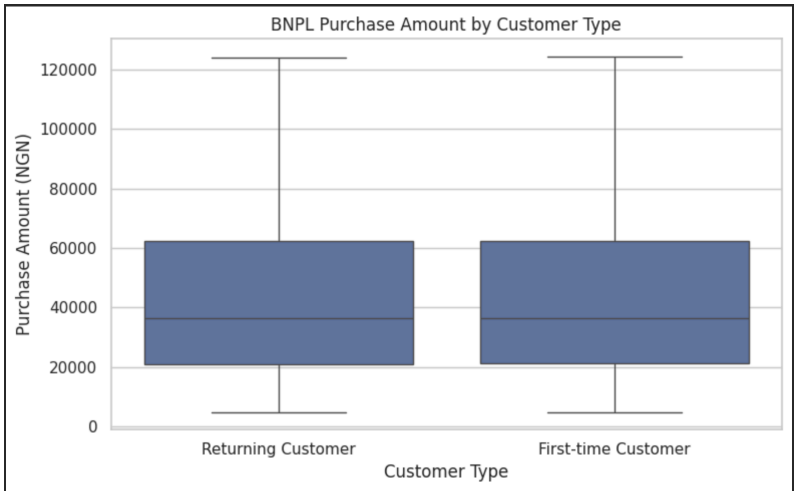
Data Visualization Strategy & Interactive Component

Our visualization strategy was designed to explore key patterns in the BNPL dataset and identify relationships that could guide the later machine learning pipeline. We focused on variables tied to repayment outcomes and borrowing behavior, including credit score, provider, purchase amount, merchant category, and customer type. For the static analysis, we used boxplots, grouped bar charts, default-rate comparisons, and a strip plot/swarm plot because these visualizations make it easier to compare repayment risk, spending patterns, and transaction behavior across meaningful customer groups. We also included an interactive chart so users can examine provider-level default patterns more flexibly than with a single static figure.

Static visualizations and findings

Question 1: Do first-time customers make different-sized purchases than returning customers?

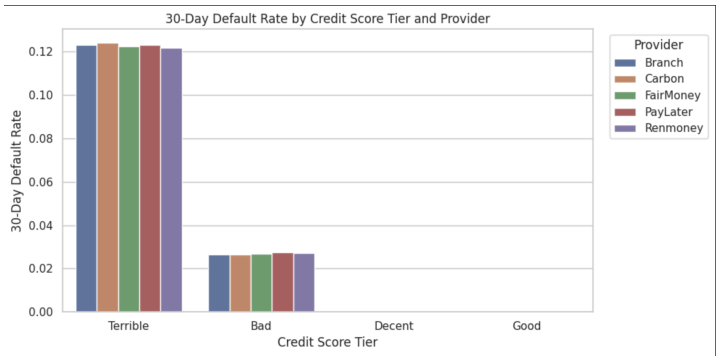
A boxplot comparing purchase amounts by customer type showed that first-time customers and returning customers have approximately similar spending distributions with first-time customers is positioned slightly higher, which suggests that their purchase amounts are slightly larger than those of returning customers. With that said, the difference is still modest, and both groups have relatively large outliers, suggesting that unusually expensive purchases is universal, not a one-certain-group thing.



Question 2: Do customers of different BNPL providers show different repayment behavior across credit score tiers?

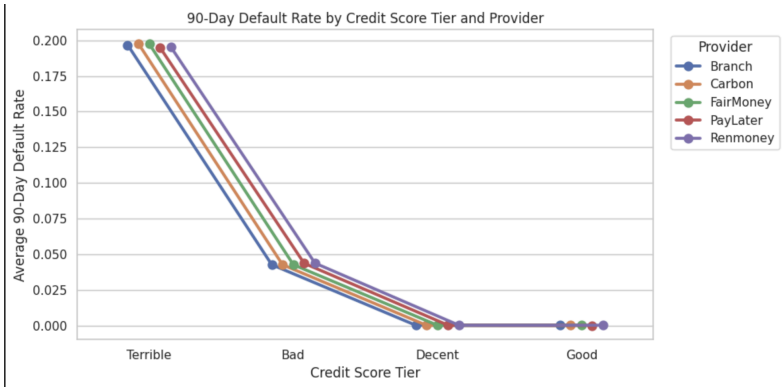
A grouped bar chart of 30-day default rate by credit score tier and provider showed that the lower the credit score tier, the higher the 30-day default rate, to such an extreme where borrowers in the Terrible tier had much higher 30-day default rates than those in the Bad tier, while the Decent and Good tiers showed little to none visible default activity. However, within the same tier, the difference of rate between

providers was insignificant, this suggests that the credit score tier is much more important than the provider in explaining short-term repayment risk.



Question 3: How does the 90-day default rate change based on credit score and provider?

Surprisingly, the longer-term pattern closely matched the short-term one when 90-day default rates were displayed across credit score tiers using the point plot, default risk dropped sharply as credit score improved. In addition, the provider curves followed nearly the same pattern, again indicating that the credit score is the strongest signal while provider contributes only modest variation.



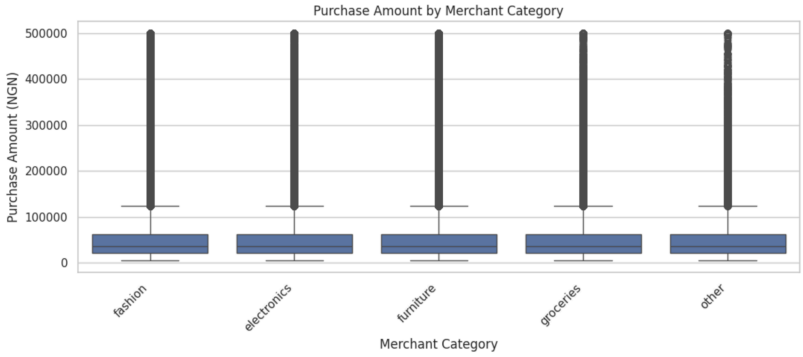
Question 4: Do customers with different credit scores behave differently?

To examine whether credit quality is also associated with borrowing behavior, we compared purchase amount across all credit score tiers. Unlike the above question where we used a boxplot, even though we had tried it here, the pattern was too hard to recognize because the data is a bit too large, the strip plot or swarm plot came in really handy here. It shows substantial overlap across tiers, with most transactions in every group concentrated at lower amounts. Some small differences appeared in the spread of purchases,

but the separation was weak overall. This concludes that credit score alone does not strongly determine purchase size, even though it is closely tied to repayment risk.

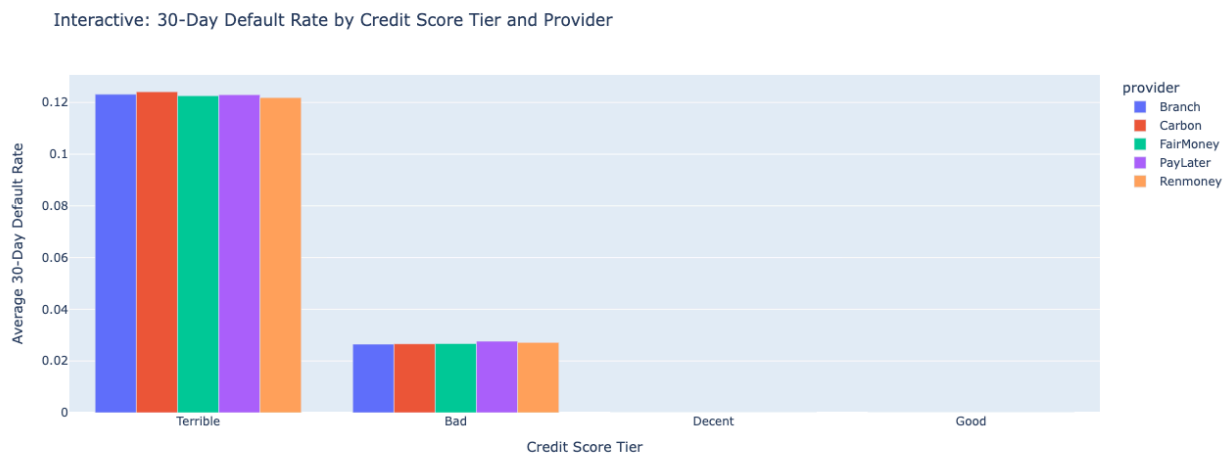
Question 5: Do different purchase categories have different spending patterns?

Merchant-category boxplots showed that purchase amount distributions are fairly similar across categories such as electronics, fashion, furniture, groceries, and other. Median purchase amounts and the main spread of the data were close across categories, and every category contained high-value outliers. This suggests that the merchant category may influence spending behavior somewhat, but the category-based differences are modest rather than extreme.



Interactive component

In addition to the static plots, we included an interactive visualization for provider-level repayment analysis. The interactive chart allows the user to examine 90-day default behavior across credit score tiers and providers by hovering over the grouped values. This makes it easier to compare providers within the same risk tier and to see how sharply default rates decline as credit scores improve. The interactive component complements the static plots by making detailed comparisons more accessible without overwhelming the page with additional figures.



(Open notebook and run the code to use the interactive portion)

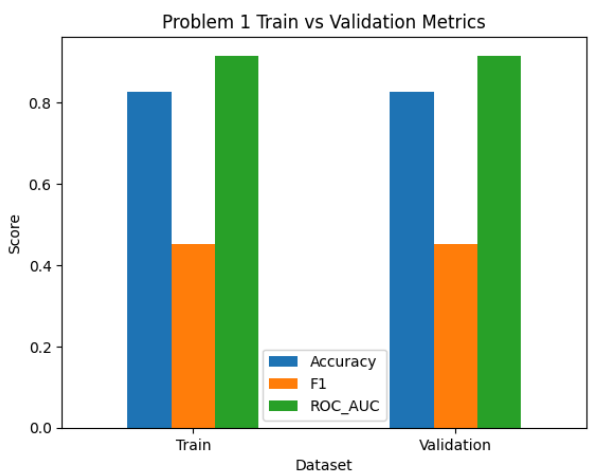
Key insights and connection to machine learning

Taken together, the visualization results show that credit score is the clearest and most consistent signal in the dataset. Lower credit tiers are associated with far higher 30-day and 90-day default rates, while provider-level differences are present but relatively small. Purchase amount, merchant category, and first-time customer status also add useful context about borrowing behavior, even when their effects are weaker than the effect of credit score. These findings help motivate the machine learning section of the project: features such as credit score, provider, principal amount, merchant category, and first-time customer status are reasonable inputs for default prediction models, and the strong separation in repayment outcomes makes classification tasks such as default_30d and default_90d especially meaningful.

Results, Code Output, and Interpretation

Machine Learning:

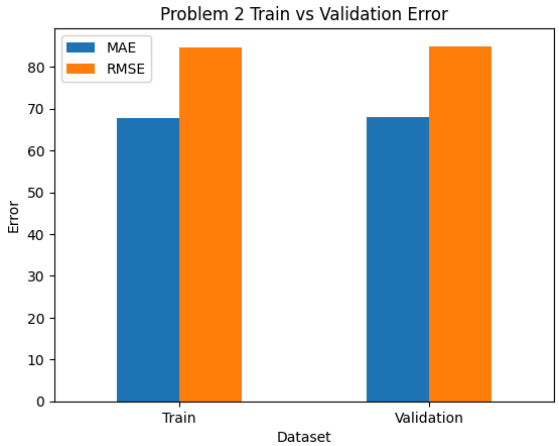
Q1: For our first model we aimed to predict whether a transaction would default within 90 days, which was done using a binary classification model. First we copied the dataset and manipulated part of the data. We converted 'purchase_date' and 'first_payment_due' into a datetime format and sorted the observations by 'purchase_date' to better reflect a time-based prediction setup you may see in the real world. From here, we ensured the data was clean without missing values, especially missing target values. Numerical columns were median imputed and standardized, categorical columns were imputed with the most frequent value and they were one-hot-encoded, although this was actually applied to hardly any columns; our dataset was already very clean. In the Numerical columns 70% of the shuffled data was used for training, 15% for validation, and 15% for the test set. Although 10% for a validation set is more traditional, predicting a default loan is risky business, and an inaccurate answer would cost a real company lots of money or a missed opportunity to make money, we want to be extra careful our model is not under or over-fitting using the val set. The final pipeline was a logistic regression model with its performance evaluated using accuracy, F1, and ROC_AUC



	Dataset	Accuracy	F1	ROC_AUC
0	Train	0.828390	0.452837	0.916471
1	Validation	0.828393	0.453517	0.916371

Q2. For our second model we aimed to predict if we could predict a customer's credit score based on loan, merchant, and transaction information we have available. Our target here was to predict 'credit_score' using a gradient boosting regression model. We chose this model because it is really good at capturing non-linear relationships between features, which makes it preferable over a simple linear baseline, considering we don't know if the relationship is perfectly straight and, honestly, we can guess that it is probably not. We applied very similar techniques such as that from problem 1, where we first copied the dataset and converted 'purchase_date' and 'first_payment' converted to datetime format then sorted chronologically by purchase_date. Using these datetime formats we were able to engineer the following features: purchase month, purchase quarter, purchase day of week Our data is already clean, so not much there, but we still ensured it was clean before splitting. Like the first model, we split using a 70/15/15 split. The final pipeline is a gradient boosting regressor model containing a random state, n_estimators=100, a learning rate of 0.05, and max depth of 3 to ensure the trees don't go too deep and

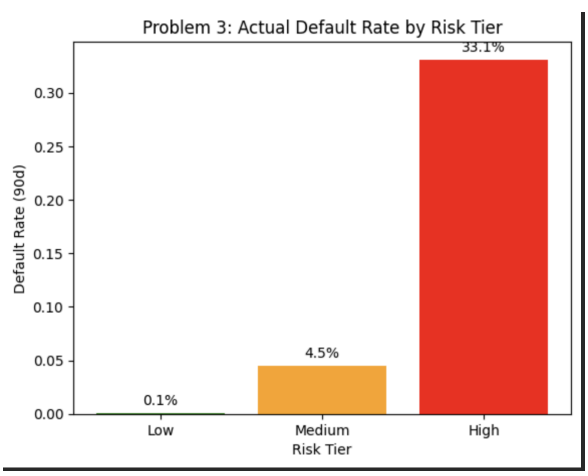
start overfitting. Five fold cross validation was performed on the training data and was evaluated using MAE, MSE, RSME, and R^2.



risk_tier	count	actual_default_rate	mean_credit_score	mean_default_prob
Low	150000	0.001	668.373	0.032
Medium	90000	0.045	595.108	0.298
High	60000	0.331	533.911	0.811

Q3. With our final machine learning model, we wanted to see if there was some way to group people into one of 3 risk categories: low, medium, or high. Customer risk segmentation was initially approached using K-Means clustering (k=3) on origination features including credit score, principal amount, interest rate, tenor, number of installments, and date-derived features. This did not end up yielding the results we were hoping for as it seemed that K-Means was splitting customers based on loan length rather than a solid risk group.

Given these limitations, risk tiers were instead derived from the predicted default probabilities produced by the Problem 1 model. Customers in the test set were segmented into three groups using percentile-based thresholds: Low risk (bottom 50%), Medium risk (50th–80th percentile), and High risk (top 20%). The revised approach produced monotonic separation across all validation metrics. Low risk customers (n=150,000) had a 0.1% actual 90-day default rate and a mean credit score of 668. Medium risk customers (n=90,000) defaulted at 4.5% with a mean credit score of 595. High risk customers (n=60,000) defaulted at 33.1% with a mean credit score of 534 and an average predicted probability of 0.811, indicating the model assigns this group with high confidence.



	Dataset	MAE	RMSE	R2
0	Train	67.719568	84.723808	0.000244
1	Validation	67.895590	84.950938	-0.000029

Citations

Dataset

Scikit Learn

Plotly

Pandas

Matplotlib